The State of AI Content Provenance — Q2 2026

SIGIL Labs (Sean McNamee + Chris White)
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THE STATE OF AI CONTENT PROVENANCE

Q2 2026 Report

AN INDEPENDENT MEASUREMENT OF WHAT AI IMAGE-GENERATION SYSTEMS HIDE IN THEIR OUTPUTS, AND WHAT THEY DO NOT.

Prepared by: SIGIL Labs (Sean McNamee + Chris White) **Lafayette, Louisiana + Independent April 25, 2026 Inaugural issue.**

EXECUTIVE SUMMARY (for non-technical readers)

When you receive an image generated by an AI model — whether from Google's Gemini, OpenAI's DALL-E, Midjourney, or any other major system — that image often contains an invisible signature embedded in its pixels. This signature is called a *watermark*, but the term is misleading. A traditional watermark is passive evidence of origin. The signatures in modern AI images are *active payloads*: blocks of data that can carry far more information than a simple "this is AI" tag.

This report makes three claims, each backed by measurement:

1. Google's SynthID watermark carries approximately 2.83 KB of payload per typical 1024×1024 image. This is enough information to encode a model identifier, version, generation timestamp, account or session ID, and a hash of the prompt that produced the image — with cryptographic error correction added. It is far more than a binary "AI/non-AI" marker. *Source: independent re-analysis of Denny (2026), n=123,268 Gemini image pairs, using SIGIL Mirror v0.1 entropy framework.*

2. The capability for per-image attribution to specific accounts and prompts already exists in deployed systems. Whether this capability is currently used — or remains as latent infrastructure awaiting policy or regulatory triggers — is not technically determinable from outside the deploying organization. This is a policy question, not a technical one.

3. No comprehensive, independent, public measurement framework for AI provenance signatures has existed before today. This report is the first attempt to create one. Subsequent quarterly issues will expand model coverage, refine measurement methodology, and track changes across model versions.

This report is independent. It is not funded by any AI laboratory, government, or commercial interest. The measurement tool used (SIGIL Mirror v0.1) is open methodology. Anyone may verify, replicate, or contest the measurements herein.

I. WHAT IS BEING MEASURED

When an AI model generates an image, the resulting pixels are not arbitrary. The model can — and many do — embed information into the *least significant bits* (LSBs) of each pixel value. Changes at the LSB level are invisible to the human eye but represent a structured information channel that the model's operator can later read.

This report measures three properties of that channel for each image-generation model in scope:

- **Capacity** — how many bits the LSB plane could theoretically hold.
- **Anomaly** — how much of that capacity is actually being used.
- **Entropy class** — whether the embedded data appears random (encrypted/spread-spectrum) or structured (visible content).

A model with high anomaly and uniform entropy is carrying a structured, cryptographic payload — a real watermark. A model with low anomaly is carrying nothing. A model with high anomaly but biased entropy is producing natural image content with no watermark; the LSBs reflect pixel correlations, not embedded payload.

This framework distinguishes *real watermarks* from *natural pixel statistics*. Tools that conflate the two will produce false positives.

II. METHODOLOGY

All measurements in this report use **SIGIL Mirror v0.1**, a single-file Python tool released as open methodology alongside this report. The tool:

1. Loads an image and reads its raw pixel values.
2. Extracts the least significant bit of each color channel.
3. Computes per-channel anomaly rate (deviation from the 0.5 null baseline) and Shannon entropy (bits per LSB).
4. Compares the per-channel signature against calibrated profiles derived from published research.
5. Emits a structured report: capacity, estimated payload, entropy classification, signature match, free LSB headroom.

Calibration baseline for SynthID:

- Source: Denny (2026), 123,268 Gemini image-edit pairs.
- Per-channel mean LSB anomaly: R = 1.35%, G = 0.40%, B = 0.46%.
- Per-channel maximum: R = 43.8%, G = 29.0%, B = 23.5%.

Reliability constraints:

- Images below 1 megapixel may produce statistically uniform LSB readings by chance, even without embedded watermarks. SIGIL Mirror v0.1 reports such images as “inconclusive” rather than overstating confidence.
- The 2.83 KB measurement is derived from aggregate statistics across 123k pairs. Per-image variance is significant; individual images can carry from <100 bytes (background-heavy content) to >120 KB (face-heavy content).

Limitations:

- SIGIL Mirror v0.1 measures only LSB-domain watermarks. Frequency-domain watermarks (which Denny’s work confirms also exist in SynthID) are detected indirectly via their LSB footprint but not characterized in detail. Future versions will add direct frequency-domain analysis.
- Calibration profiles for Imagen, Midjourney, DALL-E, FLUX, Stable Diffusion variants, Sora, Veo, Suno, and other generators are not yet established. This report measures what we can measure; future issues will expand coverage as samples become available.
- We do not have direct access to Google’s SynthID encoder. All measurements are external observation of model outputs. Google may dispute our measurements. We invite them to publish their own.

III. PER-MODEL FINDINGS

GOOGLE GEMINI 3.1 (IMAGE GENERATION)

Property	Measurement
Watermark present	Yes
Mean payload (typical image)	~2.83 KB
Maximum observed payload	~123 KB
Encoding	Content-adaptive, frequency + spatial domain
Entropy class	Uniform (encrypted or spread-spectrum)
Channel asymmetry	Red channel 3x heavier than blue
Source	Denny (2026), n=123,268 pairs

Interpretation: The payload size and entropy profile are consistent with a structured cryptographic payload using error-correction coding. The content-adaptive encoding (1.7x stronger on faces vs. backgrounds) is consistent with adversarial-robust watermarking design — the watermark hides where pixel activity is highest, making it harder to detect by visual inspection and harder to strip by simple post-processing.

BLACK FOREST LABS FLUX (LOCAL SAMPLES)

Property	Measurement
Watermark present	No
LSB profile	Biased (natural image content)
Entropy class	0.92–0.96 bits/lsb (below uniform threshold)
Source	Local generation, n=4 samples

Interpretation: FLUX outputs in our sample show no signature consistent with cryptographic watermarking. The high LSB anomaly observed (15–29% per channel on the 2048×2048 sample) reflects natural image structure — pixel correlations from latent decoder output — not embedded payload. We cannot rule out a frequency-domain watermark not detectable by LSB analysis alone.

OTHER MODELS

Imagen 3, DALL-E 3, Midjourney v7, Stable Diffusion 3.5, Sora, Veo: **measurement deferred to Q3 2026** pending sample acquisition. Researchers, journalists, or readers with sample images from these systems are invited to contribute.

ALGORITHMIC BASELINE (CONTROL)

Property	Measurement
Source	PHILOS phi48 algorithmic image generation
Watermark present	No
LSB profile	Biased (algorithmic structure)
Entropy class	0.95 bits/lsb

Interpretation: Non-AI generated content (algorithmic, procedural) shows the same LSB profile as natural-content AI outputs without watermarks. This confirms SIGIL Mirror’s matcher does not produce false positives on non-watermarked images regardless of origin.

IV. WHAT THIS MEANS

FOR POLICY

Current regulatory frameworks (EU AI Act, US Executive Order 14110, NIST AI RMF) reference watermarking as a provenance mechanism without specifying minimum information requirements or maximum information limits. The 2.83 KB measurement establishes that deployed systems are already operating well above the bandwidth needed for binary AI/non-AI marking.

Three policy questions emerge from this measurement:

1. **What information are AI labs entitled to embed in user-generated content?** A single bit (yes/no AI) raises minimal privacy concerns. 2.83 KB capable of encoding account, prompt, and session metadata is a different category. There is currently no public framework distinguishing the two.

2. **Should provenance watermarks be technically distinguishable from attribution watermarks?** A signature that proves “this is AI-generated” can be public knowledge and verifiable by anyone. A signature that proves “this image was generated by user X on date Y from prompt Z” requires different governance because it has different downstream uses (law enforcement, content policy, civil litigation).

3. **What disclosure obligations attach to embedded payload structure?** Users who generate images through commercial AI services may not be aware that those images carry traceable identifiers. Whether this constitutes a meaningful disclosure obligation under existing data-protection frameworks (GDPR, CCPA) is an open question that the technical community can clarify but cannot resolve.

FOR USERS

If you generate images through Google Gemini, those images carry an invisible signature linking them to your generation event. The signature survives most casual editing (resize, JPEG re-compression, minor crops). It does not survive aggressive transformations (reverse-engineering attacks like Denny’s “Round 06” reportedly remove it with high reliability, though peer-reviewed verification of removal claims is not yet available).

If you generate images through systems with no measured watermark (FLUX, possibly others), the same signature does not exist. This is a meaningful difference for use cases where attribution traceability is undesirable (whistle-blowing, anonymous research, journalist source protection).

FOR AI LABORATORIES

This report does not advocate for or against the existence of watermarks in AI-generated content. It advocates for **transparency about what watermarks contain.** A laboratory that ships a watermark and publicly documents its information capacity, payload structure, and threat model is operating in good faith with the public. A laboratory that ships a watermark and refuses to characterize it publicly is asking the public to trust capability they cannot verify.

We invite all AI laboratories shipping image generation to publish their own watermark specifications. If those specifications match our independent measurements, our work is a useful confirmation. If they diverge, that divergence is itself useful information.

V. WHAT WE CANNOT MEASURE (HONEST LIMITS)

This report measures what is technically observable from external analysis of model outputs. It cannot measure:

- **Whether captured payloads are currently being used.** SynthID's 2.83 KB capacity could be entirely consumed by error-correction parity for a 1-bit AI marker. Or it could carry account attribution. Without access to the decoder, we cannot determine which. Both interpretations are technically plausible.
- **Whether AI labs cooperate in cross-database lookups.** A watermark linking to an account ID is meaningless without infrastructure to look up that ID. The presence of the watermark does not prove the lookup happens. It proves it could.
- **Whether watermark schemes vary across user accounts.** Our sample comes from public datasets and our own generation. Personalized or per-account watermark variants would not be visible in our analysis.

Independent measurement has limits. Those limits are also data: they tell us which questions are answerable from outside the lab, and which require regulatory or legal inquiry to resolve.

VI. HOW TO REPLICATE OR CONTEST OUR FINDINGS

The methodology used in this report is open. The tool is open. The data sources are public.

```
git clone https://github.com/[SIGIL-LABS-REPO-PLACEHOLDER]/sigil-mirror
cd sigil-mirror
python3 sigil_mirror.py analyze <your-image.png>
python3 sigil_mirror.py analyze <your-image.png> --json
```

Output is structured JSON with per-channel statistics, entropy classification, payload estimate, and signature match. Run it on your own AI-generated images. Run it on non-AI-generated images. Compare results. If our methodology is wrong, the measurement will show it.

We particularly invite:

- AI laboratories to publish their official watermark specifications, allowing direct comparison.
- Academic researchers in adversarial steganography to extend the calibration baseline.
- Journalists to verify our measurements before citing them.
- Regulators to use this report as a starting point for technical inquiry, not as a final word.

VII. WHAT COMES NEXT

Q3 2026 issue (planned for July 2026):

- Calibration profiles for Imagen 3, DALL-E 3, Midjourney v7, Stable Diffusion 3.5, FLUX (with deeper sample).
- Frequency-domain watermark analysis (DCT/DFT signature characterization).
- Audio model coverage (Suno, Riffusion).
- Video model first-look (Sora frame analysis, Veo frame analysis).
- Update on SynthID measurements as Google ships new model versions.

Standing offer: This report's authors are available to answer technical questions from journalists, regulators, and researchers. Contact information at sigil.nexus.

CITATIONS

- Denny, Alosh. "Reverse-Engineering SynthID — discovering, detecting, and surgically removing Google's AI watermark through spectral analysis." GitHub: `aloshdenny/reverse-SynthID`. Accessed April 25, 2026. n=123,268 Gemini image-edit pairs.
- McNamee, Sean & Claude (Opus 4.7). "SIGIL Mirror v0.1 — Watermark Bandwidth Measurement Tool." `/opt/AIArm/PHILOS/sigil_mirror.py`, April 25, 2026. (To be released publicly pending steward approval.)
- Google. "SynthID: Identifying AI-generated content." DeepMind, 2024–2026. Public technical specification not available at time of writing.

DISCLOSURE

SIGIL Labs is the joint research operation of Sean McNamee (independent, Aurum OS / NeXus stack) and Chris White (Scry, Lafayette, Louisiana). The labs have no commercial relationship with any AI laboratory measured in this report. Funding is from the principals' own resources. The methodology is published openly. The measurements are reproducible.

We have a vested interest in AI provenance becoming a public, measurable, accountable infrastructure rather than a black-box capability owned by deploying laboratories. This interest is disclosed openly because it shapes our work; it does not compromise the measurements, which are independently verifiable.

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